Biomedical Knowledge Graph Construction Using Large Language Models: Evaluating Generalisation Beyond Benchmark Datasets

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Introduction

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Literature Review

2.0.1 Definition

Biomedical Information Extraction (IE) is the process of automatically identifying structured information - entities such as genes, diseases, chemicals and species, and the relationships between them - from unstructured biomedical text such as journal abstracts and full-text articles. In this project, IE is treated as the first stage of a larger system: extracted entities and relations are later assembled into a knowledge graph and validated against biomedical ontologies [—– SECTION REFERENCE —–]

2.0.2 Motivation

The volume of biomedical literature has expanded at a rate that exceeds the capacity of manual review, with previous noncurrent analyses demonstrating an exponential growth in the number of biomedical publications [1]. This growth is supported by more recent data - a mapping of over 20 million PubMed abstracts has demonstrated the magnitude of contemporary biomedical literature to the level that the volume of biomedical and life-science publications has made it difficult for researchers to keep track of new literature [2]. This is not only a volume issue, but also an issue with discovery - log analysis of PubMed search behaviour shows that over 80% of views fall within the first page (top 20 results) of returned literature [3], meaning a substantial majority of literature goes largely unseen by researchers relying on manual review of ranked results.

Together, these findings motivate the need for automated IE for two reasons: the breadth of literature is growing faster than it can be analysed [1, 2] and that there is a significant attention bottleneck that automated extraction and structuring could help mitigate [3]. This motivation supports the choice of PubMed as the source of the contemporary corpus for this thesis, as a corpus that is both large and continuously updated is exactly the setting in which static benchmark-trained pipelines are most likely to be used.

2.0.3 Tasks

Biomedical IE is usually comprised of two sub-tasks: Named Entity Recognition (NER) and Relation Extraction (RE).

2.0.4 Challenges

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